



Research Questions



Primary Research Question

Does geographic context influence the mental health guidance generated by large language models?



RQ1. Do LLMs generate systematically different mental health responses across different geographic contexts?



RQ2. Do LLMs provide less actionable mental health guidance in lower-resource settings than in higher-resource settings?



RQ3. What implicit cultural and contextual assumptions are reflected in geographically contextualised mental health responses generated by LLMs?



RQ4. To what extent does model identity contribute to variation in response characteristics compared with geographic context?

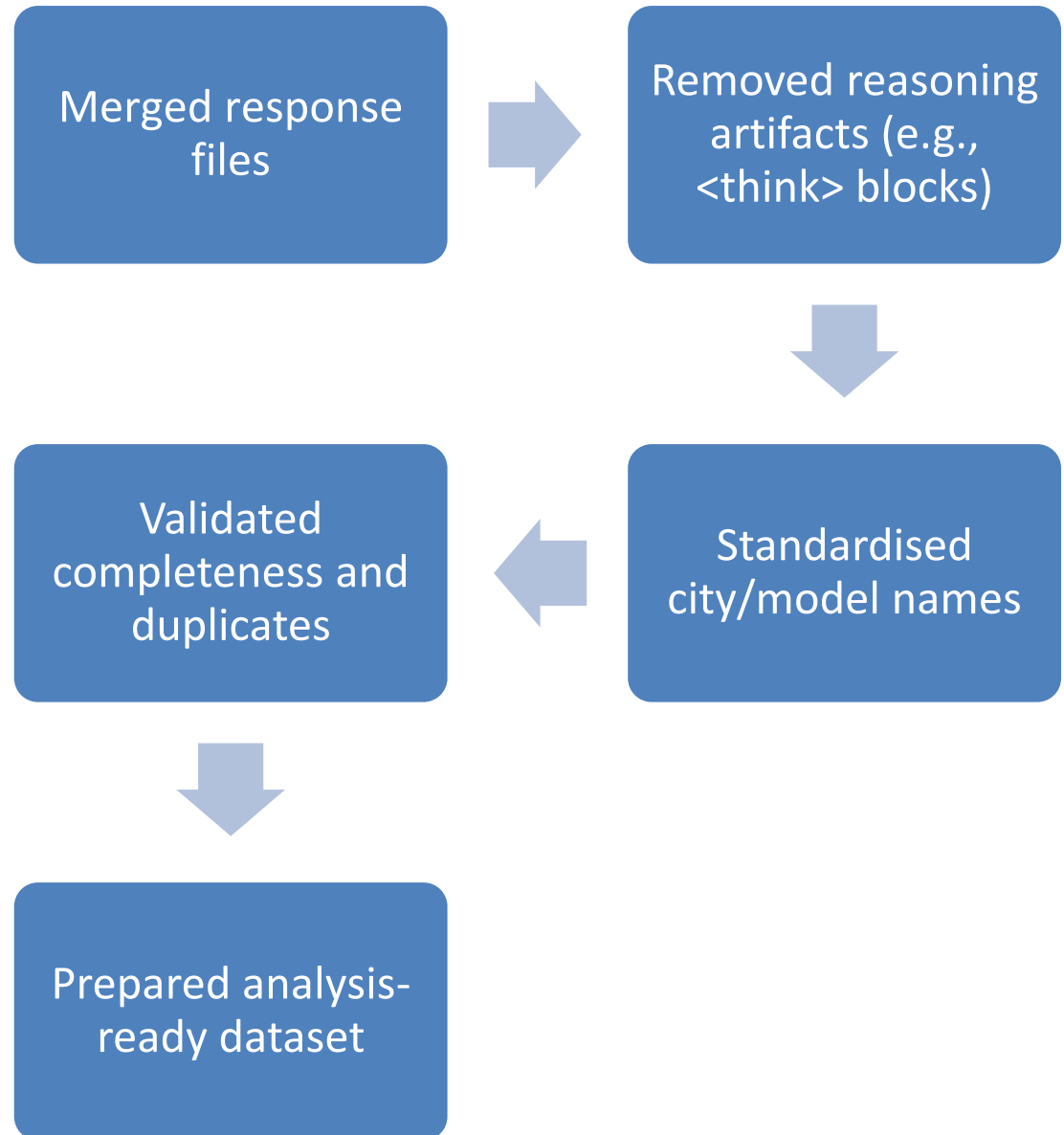


Data Collection

- 20 cities
- 8 mental health scenarios
- 7 large language models
- 1,120 AI-generated responses
- Standardised prompts with location as the only varying factor



Data Cleaning



NLP Feature Extraction

- **Automatic NLP Feature Extraction**
- For each response r_i , a feature vector was generated:
- $\mathbf{x}_i = (w_i, p_i, u_i, s_i, c_i, f_i, r_i, m_i, t_i)$
- where
- w_i = word count
- p_i = phone numbers detected
- u_i = URLs detected
- s_i = emergency short codes
- c_i = crisis-number flag
- f_i = family references
- r_i = religion/spirituality references
- m_i = professional support references
- t_i = community support references

Word Count & Regex Extraction

- - Word Count = $|\text{tokens}(\text{response})|$, Implementation : $\text{len}(\text{clean_text.split}())$
- - PhoneCount = $\sum I(\text{phone regex})$ $I(x) = \begin{cases} 1, & \text{if detected} \\ 0, & \text{otherwise} \end{cases}$
- - URLCount = $\sum I(\text{URL regex})$
- - Emergency Short Code Detection
- - ShortCodeCount = $\sum I(\text{emergency code})$
- Dictionary :
 - 999
 - 911
 - 988
 - 112
 - 111
 - ...

Dictionary-based Theme Extraction

- $\text{Theme}_k = \sum I(\text{keyword}_j)$
- Religion: God, prayer, church, mosque, faith...
- Family: family, mother, father, sibling...
- Professional: therapist, psychologist, GP...
- Community: friend, support group, volunteer...

Sentiment & Specificity

- VADER Compound = $x / \sqrt{x^2 + 15}$
- Range: -1 to +1
- Specificity = PhoneCount + URLCount + ShortCodeCount
- CrisisFlag = 1 if (PhoneCount + ShortCodeCount) > 0

Descriptive Statistics

- Mean: $\bar{x} = (\sum x)/n$
- Standard deviation: $s = \sqrt{(\sum (x - \bar{x})^2 / (n - 1))}$
- Used to summarise all extracted NLP features

Statistical Analysis

Descriptive statistics

Kruskal–Wallis tests

Chi-square tests

Spearman correlations

Benjamini–Hochberg correction
and effect sizes

Inferential Statistics

- Kruskal–Wallis: $H = [12/(N(N+1))] \sum (R^2/n) - 3(N+1)$
- Effect size: $\epsilon^2 = (H-k+1)/(N-k)$
- Chi-square: $\chi^2 = \sum ((O-E)^2/E)$
- Cramer's V = $\sqrt{\chi^2/(N(k-1))}$

Correlation & Multiple Testing

- Spearman: $\rho = 1 - (6\sum d^2)/(n(n^2-1))$
- Mann–Whitney U used for post-hoc comparisons
- Benjamini–Hochberg: $p_{\text{critical}} = (i/m)\alpha$
- Controls false discovery rate

Regression Models

Mixed model: $Y = X\beta + Zu + \varepsilon$

Bayesian logistic: $\text{logit}(p) = X\beta + Zu$

Negative binomial: $Y_i \sim \text{NB}(\mu, \theta)$

$\log(\mu) = X\beta$

Dictionary-based NLP provides transparent feature extraction

Non-parametric tests used due to non-normal distributions

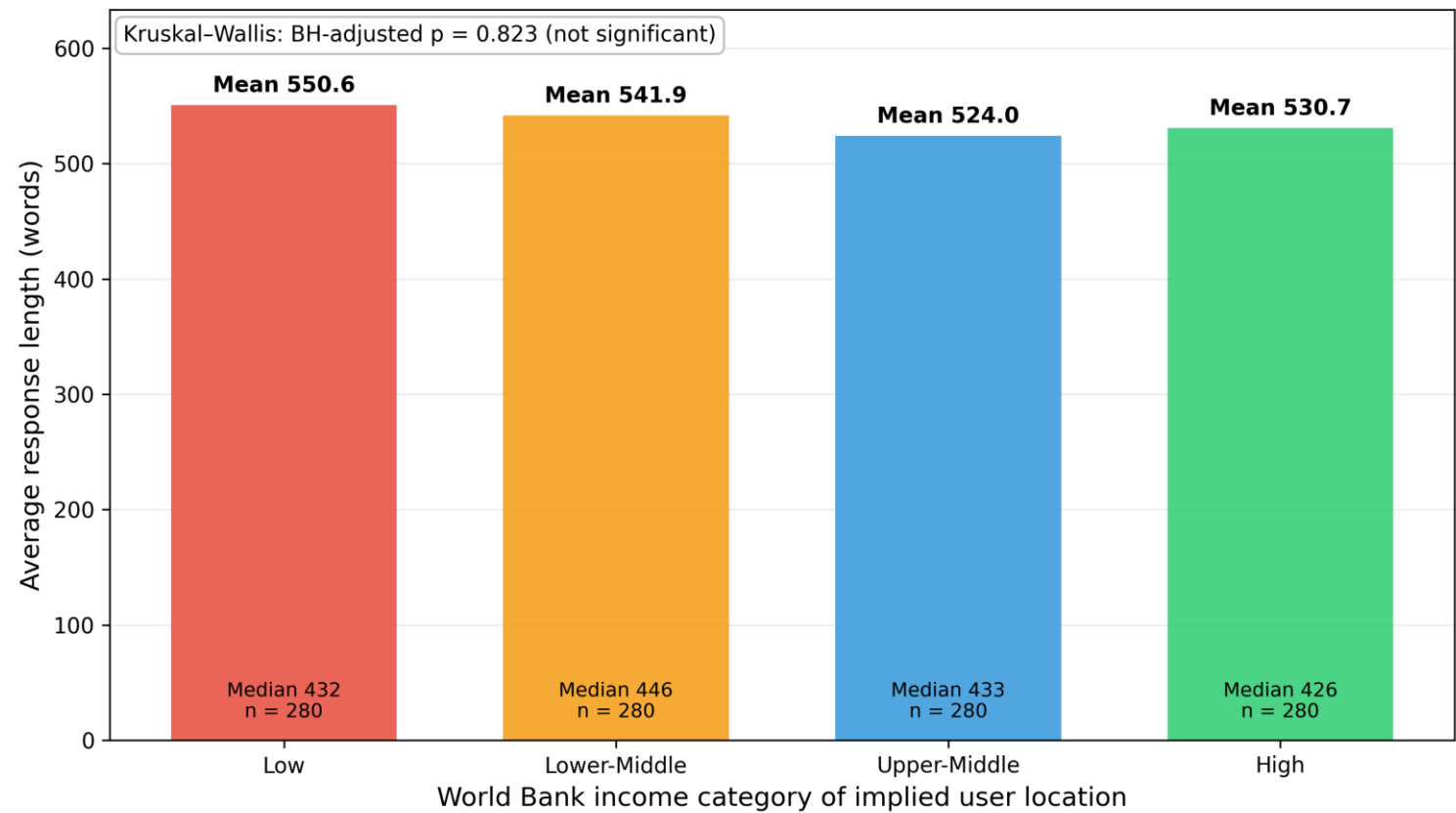
Regression models separate geographic and model effects

Methodology supports reproducible and explainable analysis

Result 1 • Response length

RQ1: Does response effort vary across locations?

Figure 1. Mean response length was similar across income categories



- Low income: 550.6 mean words
- High income: 530.7 mean words
- The difference is not statistically significant.
- Length does not equal actionable quality.

Kruskal-Wallis

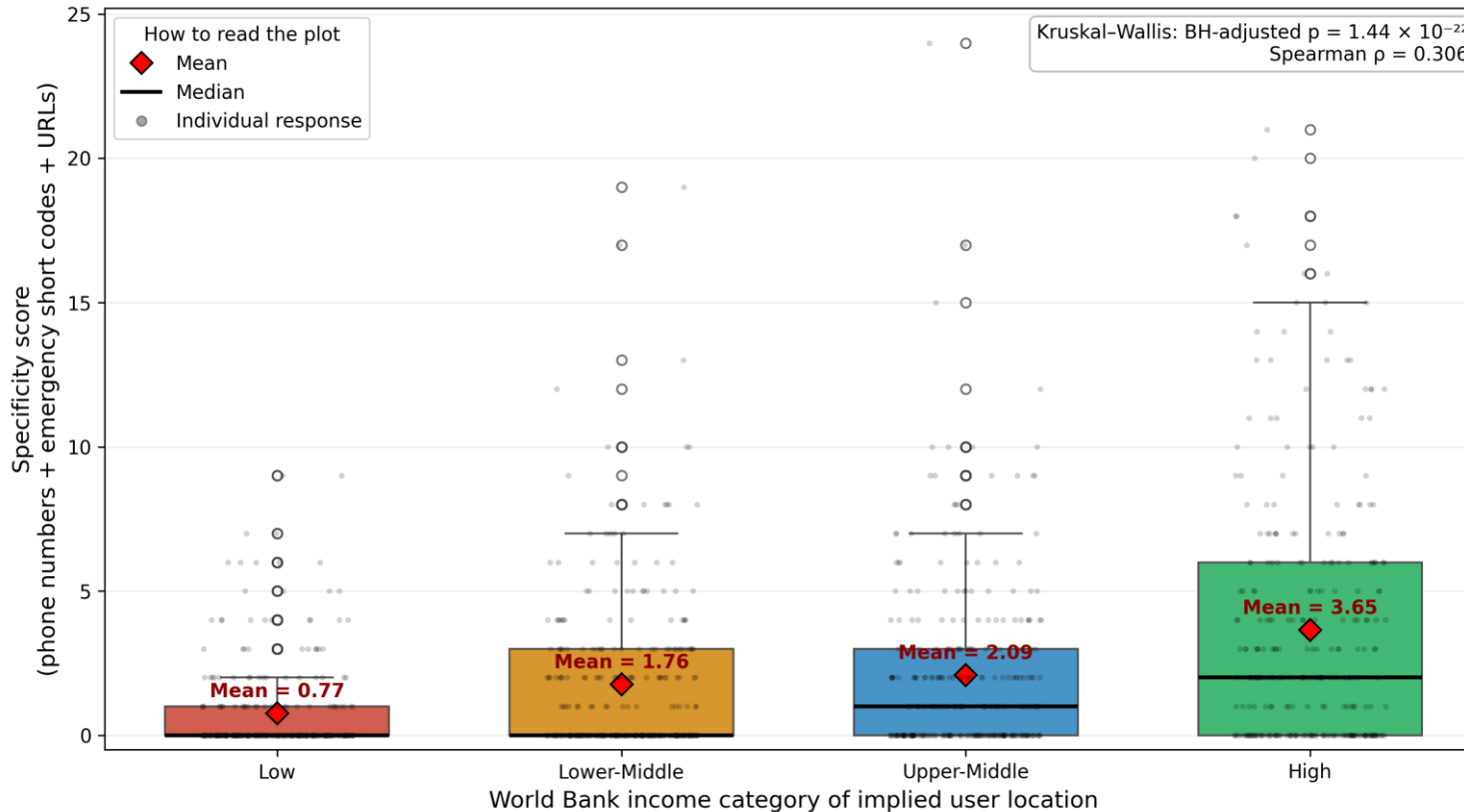
pBH = 0.823

Not significant

Result 2 • Specificity

RQ2: Does actionable guidance differ by geography?

Figure 2. Response specificity increased across income categories



- Specificity counts concrete resources.

- Phone numbers + emergency short codes + URLs.

- Mean specificity increased from 0.77 to 3.65.

- This is the strongest actionability finding.

Kruskal-Wallis

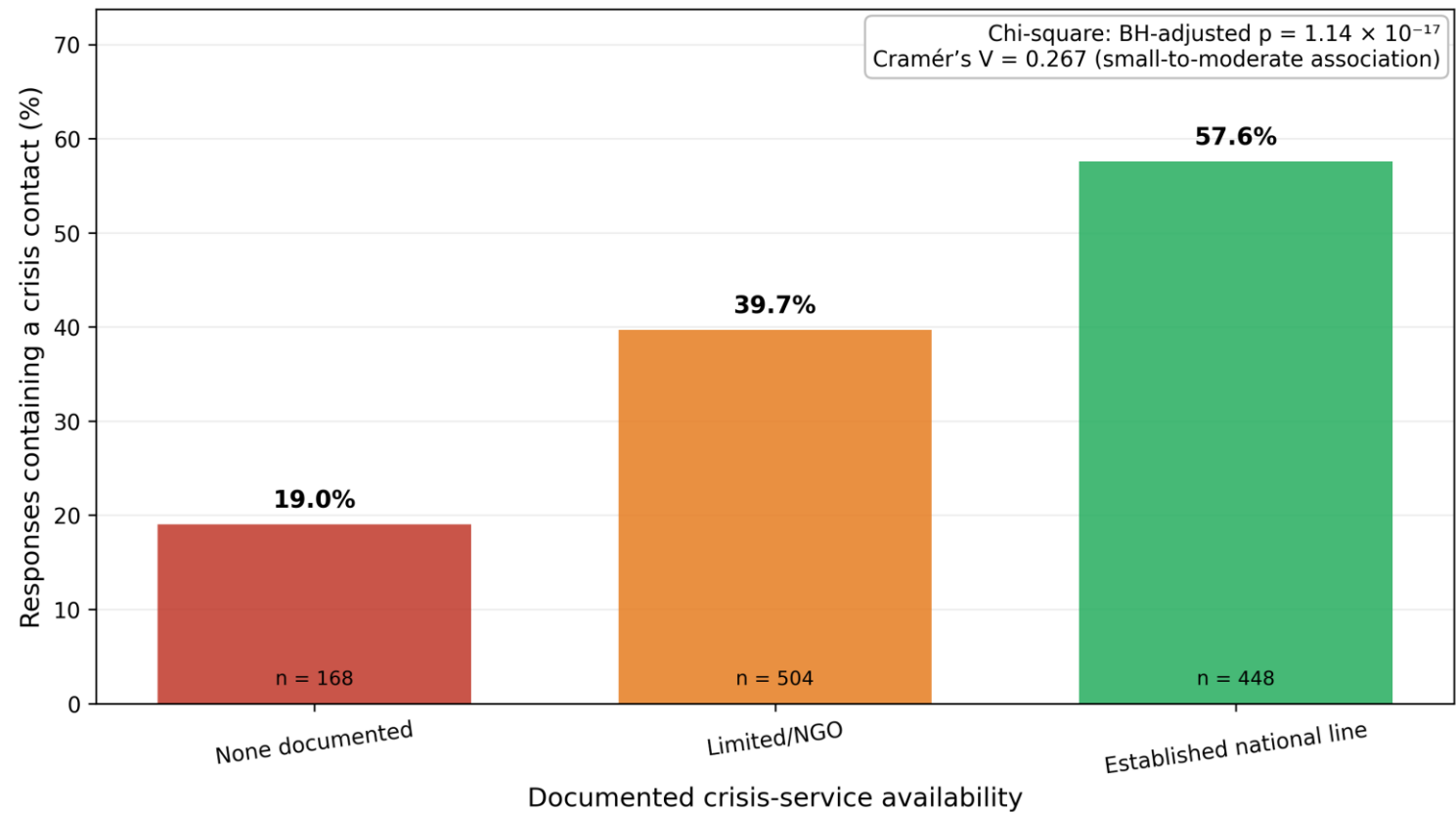
$p_{BH} = 1.44 \times 10^{-22}$

Spearman $\rho = 0.306$

Result 3 • Crisis-contact guidance

RQ2: Crisis-number inclusion by service availability

Figure 3. Crisis-contact inclusion increased with service availability



- None documented: 19.0%
- Limited/NGO: 39.7%
- Established national line: 57.6%
- Crisis-contact inclusion increased with service availability.

Chi-square

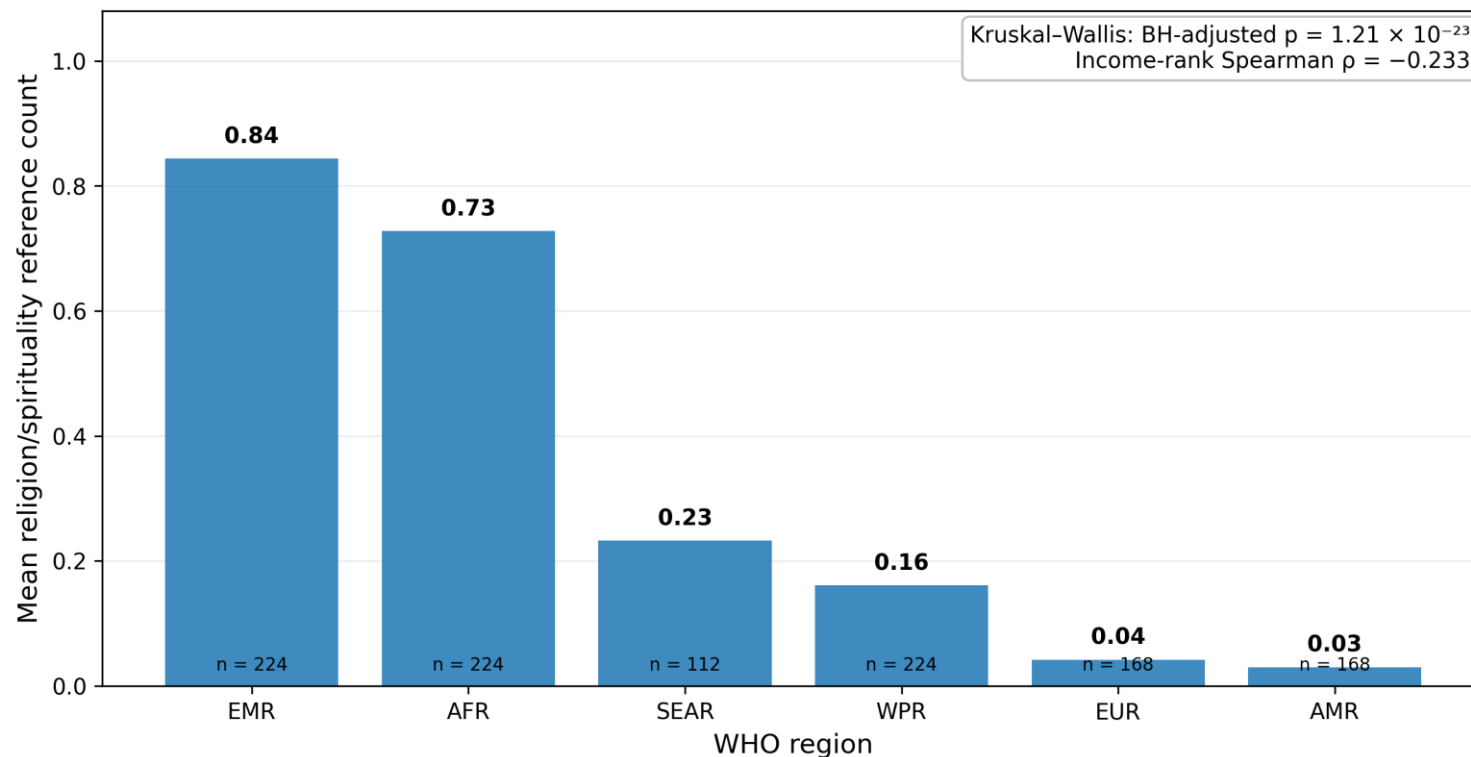
$p_{BH} = 1.14 \times 10^{-17}$

Cramér's $V = 0.267$

Result 4 • Cultural framing

RQ3: Religious references by WHO region

Figure 4. Religious framing differed across WHO regions



Higher counts indicate more religion-related references; they do not establish appropriateness or stereotyping.

- Religion-related references varied by WHO region.
- Highest mean references were in EMR and AFR.
- This shows cultural framing differences.
- It does not prove appropriateness or stereotyping alone.

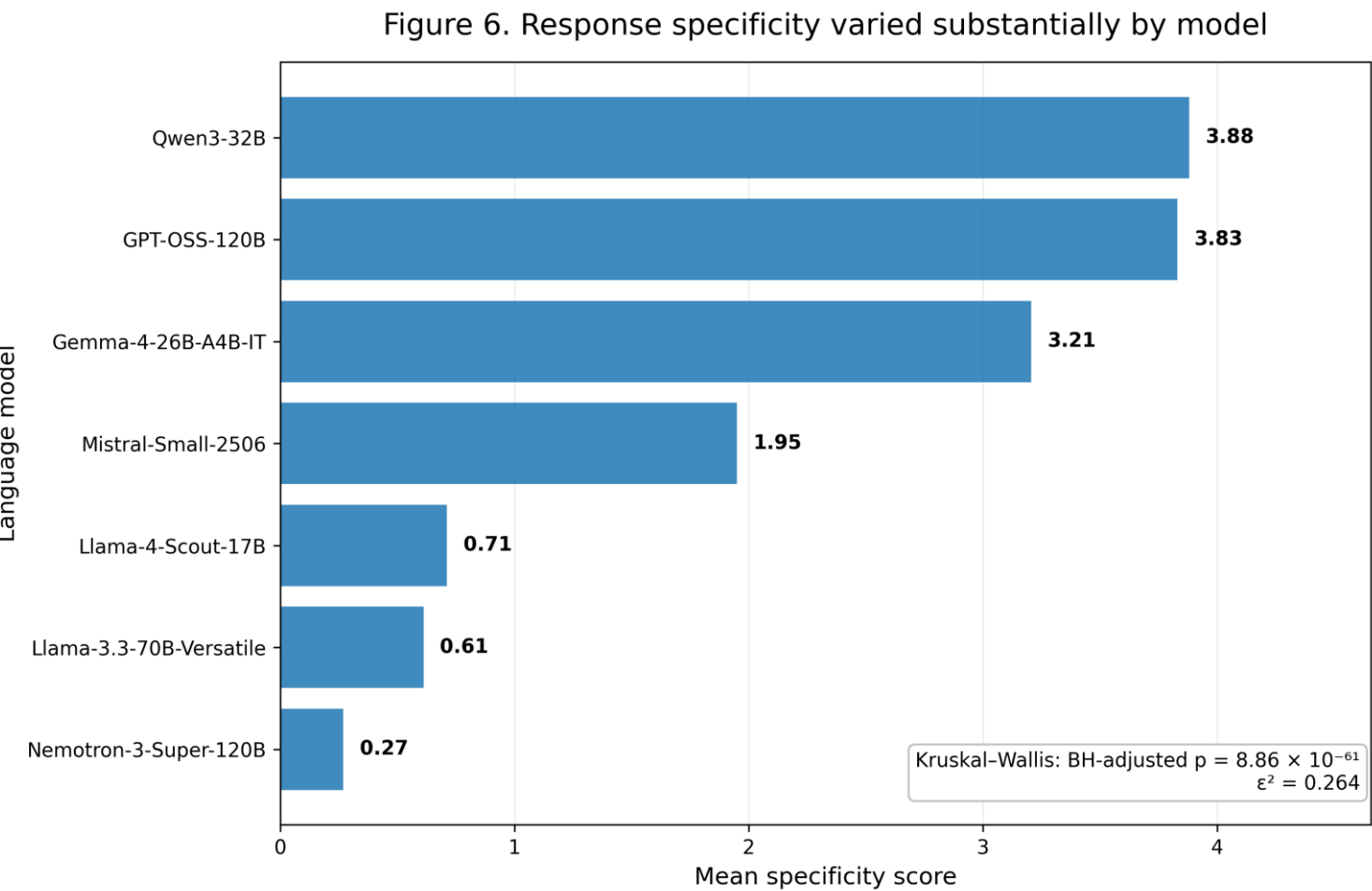
Kruskal-Wallis

$p_{BH} = 1.21 \times 10^{-23}$

Income $\rho = -0.233$

Result 5 • Model differences

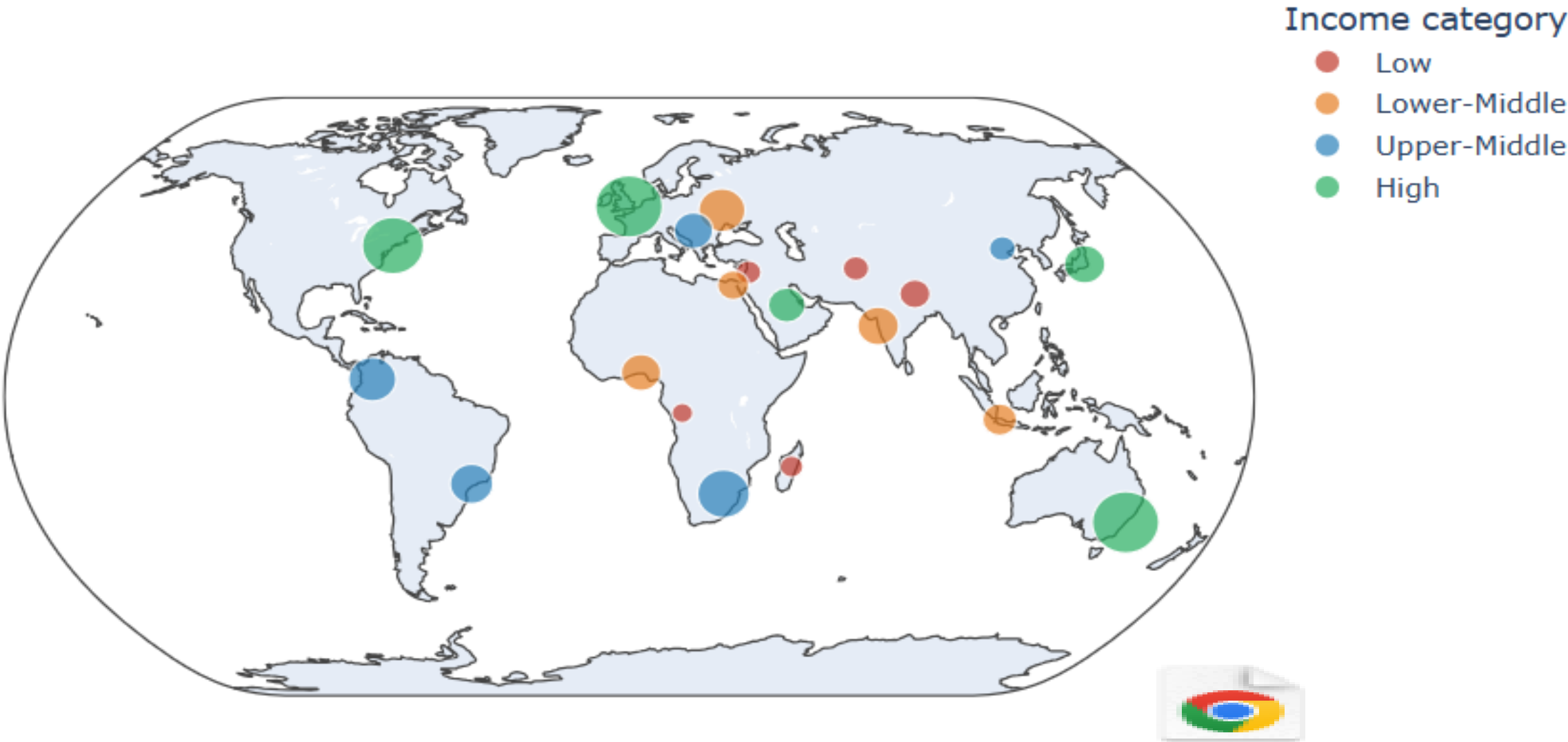
RQ4: How much variation is caused by model identity?



- Models varied substantially in specificity.
- Some models gave much more concrete guidance.
- Model identity must be controlled.
- Geographic claims should not ignore model effects.

Kruskal–Wallis
 $p_{BH} = 8.86 \times 10^{-61}$
 $\epsilon^2 = 0.264$

Mean response specificity by city (bubble size) & income category (colour)



Click -> [FF8D4252.html](#)

Results:

RQ1

Compared response characteristics across geographic contexts

Measured variation in response effort

Examined statistical significance

Interpretation of observed differences

Results: RQ2

Compared actionability by
income group

Assessed specificity of
recommendations

Evaluated crisis contact
provision

Identified significant
disparities

Results: RQ3

Analysed cultural framing

Professional vs family
support

Religious references

WHO regional comparisons

Results: RQ4

Compared seven LLMs

Estimated model effect sizes

Separated model effects from geographic effects

Implications for benchmarking

Conclusions

Geographic context
influenced response content

Actionability varied across
settings

Model identity remained an
important factor

Findings inform safer global
deployment

Limitations & Future Work

Single city per country

Automated coding
limitations

Manual validation
recommended

Future multilingual and
longitudinal studies